Laplaceova transformacija

~ kontinuirani signali ~

→ Dvostrana, ide od $-\infty$ do $+\infty$

↳ mi ćemo definirati samo JEDNOSTRANU ⇒ POBUDA i ODZIV moraju biti KAUZALNI!

$$X(s) = \int_{0^-}^{\infty} x(t)\, e^{-st}\, dt$$ UNILATERALNA L-transf.

$$\delta(t) \longrightarrow \int_{0^-}^{\infty} \delta(t) e^{-st} dt = e^0 = 1$$

$$\cos(\omega_0 t)\cdot\mu(t) \longrightarrow \frac{s}{s^2+\omega_0^2}$$

$$\delta(t-t_0) \longrightarrow e^{-st_0}$$

$$\sin(\omega_0 t)\cdot\mu(t) \longrightarrow \frac{\omega_0}{s^2+\omega_0^2}$$

$$\mu(t) \longrightarrow \int_{0^-}^{\infty} e^{-st} dt = \frac{e^{-st}}{-s}\Big|_{0^-}^{\infty} = \frac{1}{s}$$

$$e^{-at}\mu(t) \longrightarrow \frac{1}{s+a}$$

1. $w(t) = a\,x(t) + b\,y(t) \longrightarrow W(s) = a\,X(s) + b\,Y(s)$

2. Pomak $\quad x(t-t_0)\,\mu(t-t_0) \longrightarrow e^{-st_0}\cdot X(s)$

3. Frekvencijski pomak $\quad x(t)\cdot e^{s_0 t} \longrightarrow X(s-s_0)$

4. Kompresija $\quad x(\alpha t) \longrightarrow \frac{1}{\alpha}\, X\left(\frac{s}{\alpha}\right)$

5. Konvolucija

$x_1(t)\,\mu(t)$
$x_2(t)\,\mu(t)$ $\Rightarrow y(t) = x_1(t) * x_2(t) \longrightarrow Y(s) = X_1(s)\cdot X_2(s)$

6. Derivacija

$$x'(t) \longrightarrow s\,X(s) - x(0^-)$$

$$x''(t) \longrightarrow s^2 X(s) - s\cdot x(0^-) - x'(0^-)$$

$$x^{(k)}(t) \longrightarrow s^k X(s) - s^{k-1} x(0^-) - s^{k-2}\cdot x^{2}(0^-) - \dots - x^{(k-1)}(0^-)$$

7. Integracija

$$x(t) \longrightarrow X(s)$$

$$\int_{0^-}^{t} x(\tau)\, d\tau \longrightarrow \frac{1}{s}\, X(s)$$

8. Derivacija u frekv.

$$-t\cdot x(t) \longrightarrow X'(s)$$

$$(-t)^k x(t) \longrightarrow X^k(s)$$

Primjer: $y''(t) + 2y'(t) + 5y(t) = u(t)$

$u(t) = 5\mu(t)$

$y(0^-) = -1 \qquad y'(0^-) = -1$

* ni kod Laplacea ne moramo preračunavati početne uvjete

$$u(t) \longrightarrow U(s) = 5 \cdot \frac{1}{s}$$

$$s^2 Y(s) - s y(0^-) - y'(0^-) + 2\left[sY(s) - y(0^-)\right] + 5Y(s) = U(s)$$

$$Y(s)\left[s^2 + 2s + 5\right] - s y(0^-) - y'(0^-) - 2y(0^-) = U(s)$$

$$Y(s) = \underbrace{U(s) \cdot \boxed{\frac{1}{s^2+2s+5}}}_{\text{MIRNI}} + \underbrace{\frac{s y(0^-) + 2y(0^-) + y'(0^-)}{s^2+2s+5}}_{\text{NEPOBUĐENI}}$$

prijenosna fja: $H(s) = \dfrac{Y(s)}{U(s)}$

Mirni:

$$Y_m(s) = \frac{1}{s^2+2s+5} \cdot \frac{5}{s} \longrightarrow s_{1,2} = -1 \pm 2j$$

, kod Laplacea ne moramo faktorizirati jer imamo transformacije za cos i sin

$$Y_m(s) = \frac{As+B}{s^2+2s+5} + \frac{C}{s} \longrightarrow As^2 + Bs + C(s^2+2s+5) = 5$$

$$A + C = 0 \qquad B + 2C = 0 \qquad C5 = 5$$

$$C = 1 \qquad A = -1 \qquad B = -2$$

$$\boxed{Y_m(s) = \frac{-s-2}{s^2+2s+5} + \frac{1}{s}} = \frac{-s-2}{(s+1)^2+4} + \frac{1}{s} = -\frac{s+1}{(s+1)^2+4} - \frac{1}{(s+1)^2+4} + \frac{1}{s}$$

$$= -\frac{s+1}{(s+1)^2+4} - \frac{1}{2} \cdot \frac{2}{(s+1)^2+4} + \frac{1}{s} \Rightarrow Y(s)$$

$$\boxed{y_m(t) = \left[-e^{-t} \cdot \cos(2t) - \frac{1}{2} e^{-t} \sin(2t) + 1\right] \mu(t)}$$

Odziv nepobuđenog:

$$Y_0(s) = \frac{sy(0^-) + 2y(0^-) + y'(0^-)}{s^2+2s+5}$$ uvrstimo poč. uvjete $$= \frac{-s-2-1}{s^2+2s+5} =$$

$$Y_0(s) = \frac{-s-2-1}{(s+1)^2+4} = -\frac{s+1}{(s+1)^2+4} - \frac{2}{(s+1)^2+4}$$

$$\boxed{y_0(t) = \left[-e^{-t}\cdot\cos(2t) - e^{-t}\cdot\sin(2t)\right]\cdot\mu(t)}$$

Totalni odziv: $y(t) = y_0 + y_m$

$$\boxed{y(t) = \left[-2e^{-t}\cdot\cos(2t) - \frac{3}{2}e^{-t}\sin(2t) + 1\right]\mu(t)}$$

U slučaju kompleksnih

Z

→ kompleksne uvrsti kompleksnost

L

→ kaj. kompl → ostavi i računaj cos i sin

→ obični (realni) ⟹ rastavi na polovi